Lab 4

Visualizing Population Flow

Aggregate and PUMA flow data, flow visualization tools

DATA 20195 Spring 2026

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American Community Survey (ACS)

This is our third week looking at the ACS! In response to some questions we've gotten in previous labs, here is some more information about the survey.

- U.S. Census Bureau contacts over 3.5 million households per year, approximately 295,000 per month, by mail to notify that they have been selected to participate
  - mailed notice provides instructions for completing the survey online
  - paper questionnaire is later mailed for those that do not respond online
  - in-person visits by Census employees occur hard to reach addresses, such as Remote Alaska, or for those who could not be reached via Internet or paper questionnaire
- Each address has about a 1-in-480 chance of being selected in a month, 1-in-38 per year, and no address should be selected more than once every 5 years.

https://www.census.gov/programs-surveys/acs/respond/other-ways-to-respond.html#:~:text=Every%20year%2C%20the%20U.S.%20Census,out%20the%20form%20in%20person
https://www.census.gov/content/dam/Census/programs-surveys/acs/about/ACS_Information_Guide.pdf .

ACS Migration Data

Person 1 (continued)

F *Answer question 12 if this person has a bachelor's degree or higher. Otherwise, SKIP to question 13.*

12 **This question focuses on this person's BACHELOR'S DEGREE. Please print below the specific major(s) of any BACHELOR'S DEGREES this person has received.** *(For example: chemical engineering, elementary teacher education, organizational psychology)*

13 **What is this person's ancestry or ethnic origin?**

(For example: Italian, Jamaican, African Am., Cambodian, Cape Verdean, Norwegian, Dominican, French Canadian, Haitian, Korean, Lebanese, Polish, Nigerian, Mexican, Taiwanese, Ukrainian, and so on.)

14 **a. Does this person speak a language other than English at home?**

☐ Yes

15 **a. Did this person live in this house or apartment 1 year ago?**

☐ Person is under 1 year old ➔ *SKIP to question 16*

☐ Yes, this house ➔ *SKIP to question 16*

☐ No, outside the United States and Puerto Rico – *Print name of foreign country, or U.S. Virgin Islands, Guam, etc., below; then SKIP to question 16*

☐ No, different house in the United States or Puerto Rico

b. Where did this person live 1 year ago?

Address (Number and street name)

Name of city, town, or post office

Name of U.S. county or municipio in Puerto Rico

Name of U.S. state or Puerto Rico

ZIP Code

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International Flows

County to County flows

Screenshot from the 2020 Census Questionaire

ACS Migration Data

2 levels of indices

estimate

margin of error

a_fips	b_fips	a_state	a_county	b_state	b_county	b_to_a_flow_est	b_to_a_flow_moe	a_to_b_flow_est	a_to_b_moe	net_flow_est	net_flow_moe	gross_flow_est	gross_flow_moe
36005	36047	New York	Bronx County	New York	Kings County	6924	969	4040	716	2884	1269	10964	1136
	36061	New York	Bronx County	New York	New York County	11469	1191	4965	883	6504	1490	16434	1475
	36081	New York	Bronx County	New York	Queens County	5242	978	3701	628	1541	1108	8943	1214
	36085	New York	Bronx County	New York	Richmond County	461	183	267	166	194	260	728	234
36047	36005	New York	Kings County	New York	Bronx County	4040	716	6924	969	-2884	1269	10964	1136
	36061	New York	Kings County	New York	New York County	21743	1419	12241	1085	9502	1912	33984	1652
	36081	New York	Kings County	New York	Queens County	8942	889	14949	1381	-6007	1752	23891	1526
	36085	New York	Kings County	New York	Richmond County	1038	254	6065	1014	-5027	1029	7103	1061

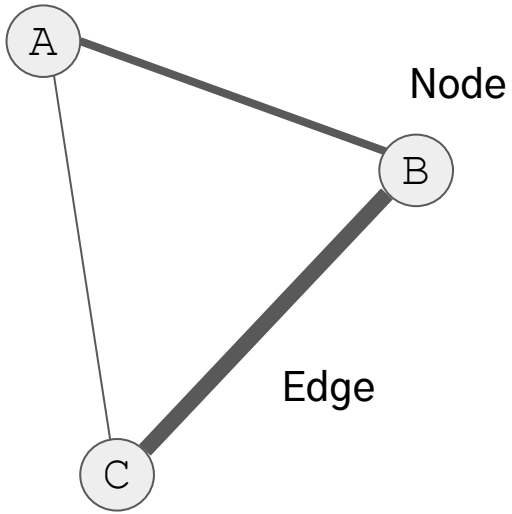
What does a negative net flow mean?

"Net flow from B to A"

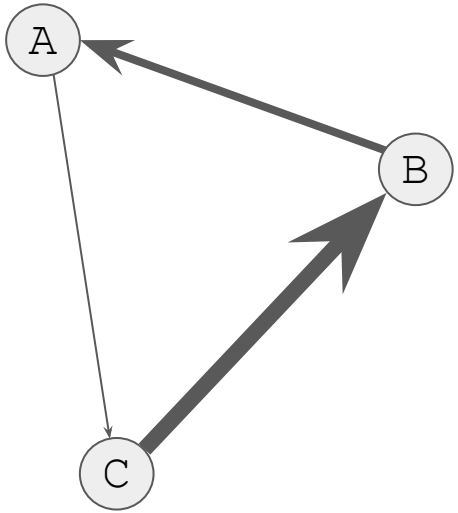
What does the gross flow mean?

Graph Basics

Undirected Graph



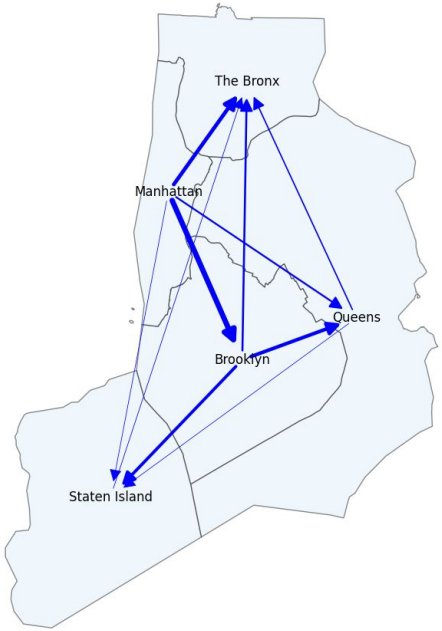
Directed Graph



In this lab, we will use a Python package called `NetworkX` to visualize graphs.

Net or both ways?

Average annual net inter-borough migration, 2016–2020 (ACS)

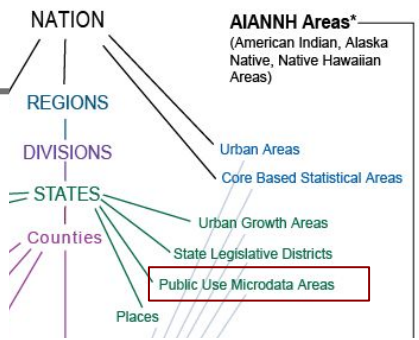


We chose the "**net**" graph -- flows go both ways, but we took the difference and plotted only one arrow.

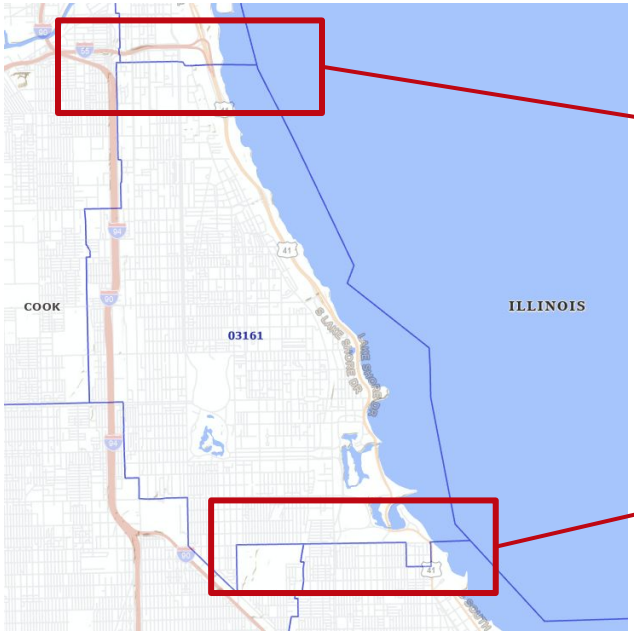
How might you go about conceptualizing the viz for showing **BOTH** flows?

Why might you prefer one or the other?

Public Use Microdata Areas (PUMAs)



- Individual-level census data.
- At least 10,000 people in each PUMA.
- Trades off geographical resolution for data granularity.



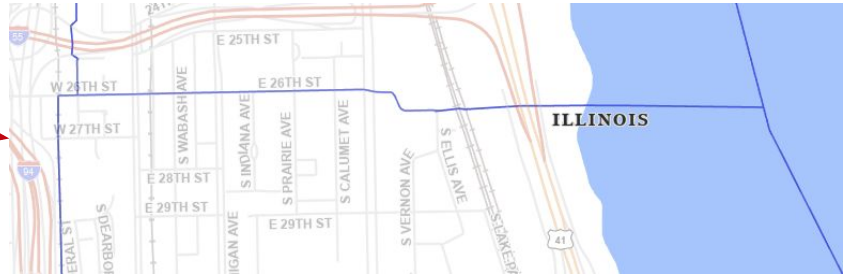
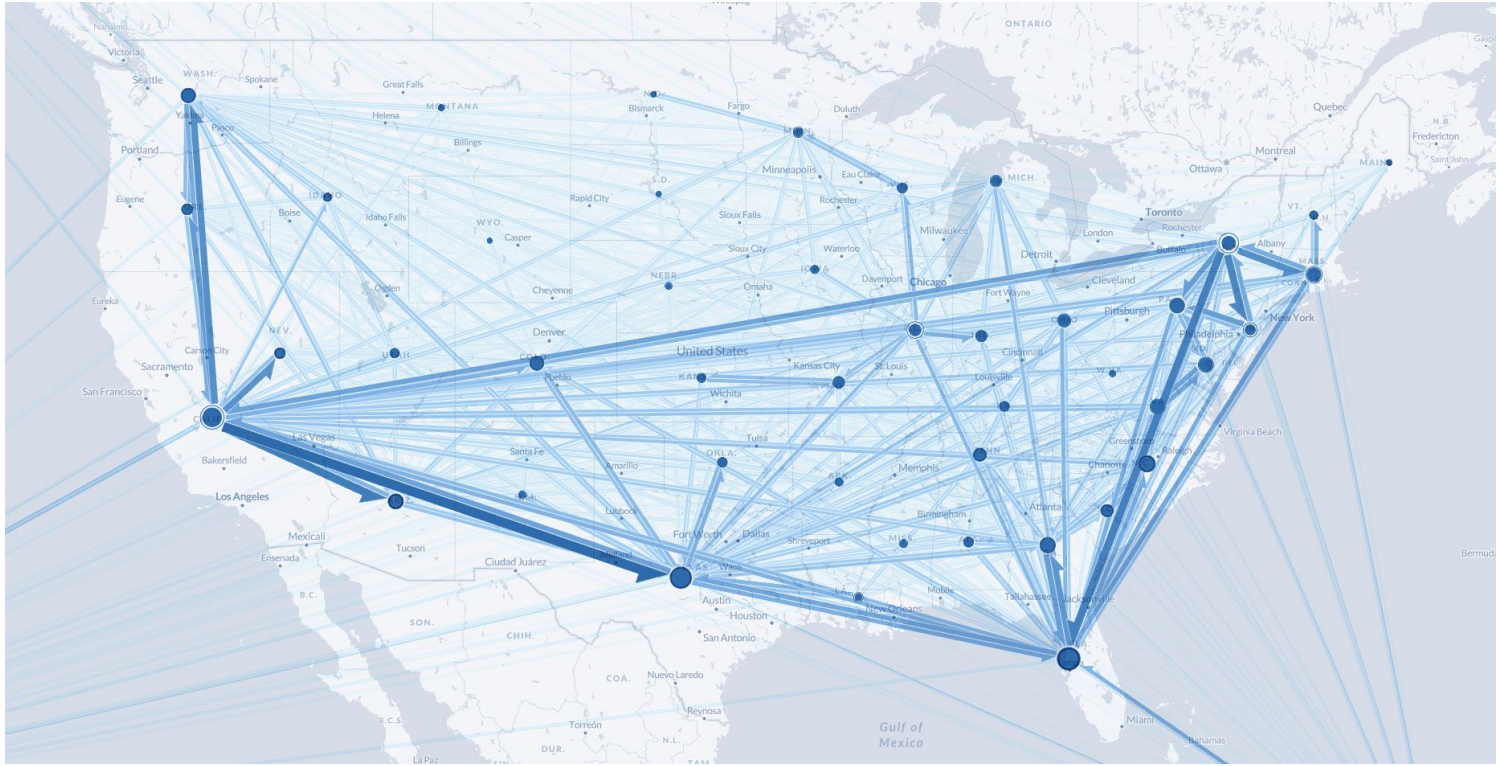


Image Credit: TIGERweb

Interactive flow visualization with `FlowmapBlue`



Trends? Observations? Comments?